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DEPARTMENT OF PSYCHOLOGY

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ABSTRACTS

1. P. Jusczyk - Infants' Discrimination of the Relative Onset-Time of Two-Component Tones.

A great deal of speech perception research has focused upon the nature of voice-onset-time differences in stop consonants. Yet, the nature of the mechanisms responsible for the perception of these differences is still the subject of much debate. The present study examines the infant's perception of a cue suggested as a possible psycho-acoustic basis for the perception of voice-onset-time differences.

2. R. Pinter & L. Harris - Spatial and Temporal Tuning Characteristics of Cat Superior Colliculus.

We have measured the response of visual cells in the cat's superior colliculus to moving gratings of sinusoidal luminance profile. Spatial frequency (cycles/deg), velocity (degs/sec) and contrast were varied and presentations were randomly interleaved under computer control. Some cells preferred a particular temporal frequency while many responded best at a certain spatial frequency and velocity.

3. A. Fröhlich & I. Meinertzhagen - Synaptogenesis in the Fly's Visual System.

Rules which control the formation or selection of synapses during development are the last step in a chain of cellular control mechanisms by which neurons interconnect with great specificity. To study the nature of these rules we have examined anatomically populations of synapses in the first optic neuropile, or lamina, of the fly *Musca* in the adult and in various immature stages. For each has been scored the number of synapses and the identity of elements communicating synaptically. Synapses have been scored from limited series of up to 50 consecutive micrographs prepared from material fixed and processed conventionally for electron microscopy. We have found that synapses form with great specificity. In the adult, the first-order synapse is an elongate tetrad with dyadic symmetry. The formation of these synapses occurs relatively late in development (50% - 80% of pupal life) after the neural elements are represented in their final positions, but before the surrounding glial cell processes have ensheathed the elements. Some characteristic features of synaptic ultrastructure are however acquired between 80% - 100% of pupal life. These include the platform component of a conspicuous T-shaped presynaptic density, a subsynaptic cisterna in each of two (L1 and L2) postsynaptic elements, the full complement of synaptic vesicles and the presence of nearby glial invaginations (capitate projections).

4. T. Whalen - Short-Term Memory in Pigeons.

One method of assessing short-term memory in pigeons requires that the characteristics of a stimulus be recalled. Another method allows the formation of an intention to respond which may be recalled in lieu of the stimulus. The present research demonstrates that pigeons must have prior training with delays if they are required to recall the stimulus, but remembering is equally high if an intention to respond could be produced, regardless of prior training.

5. G. Goddard - Mixed Excitatory Inhibition Control of Dentate Granule Cells by Hippocampal Commissure Activation.

Douglas showed that synaptic enhancement of perforant path inputs to the hippocampus are aborted by high-frequency activation of the hippocampal commissure - a pathway thought to be excitatory but which appeared in our studies to be inhibitory. Application of Bicuculline has revealed that the pathway is a mixture of both.

6. J. Gardner - Binocular Interactions in Cat Visual Cortex.

In cat visual cortex, many cells can be driven through one eye only. Other cells can be driven equally through the two eyes. One would think that neurons which receive excitatory input from only one eye were important for monocular viewing while those receiving equal input from the two eyes were important for binocular vision. The data however, suggest otherwise.

7. J. Barresi & B. MacCallum - Parsing Principles for Patterns.

In several experiments subjects were asked to partition dot patterns into their "natural" parts. Although the number of parts perceived in a given pattern varied across subjects, most of their partitions could be fit into a single nested hierarchy of parts. It is suggested that a uniform set of parsing principles which generates such hierarchies may form the basis of pattern recognition processes.

8. L. Harris & M. Cynader - Abnormalities in the Vestibulo-ocular Reflex and Optokinetic Nystagmus of Dark-Reared Cats.

The effectiveness of the vestibulo-ocular reflex of dark-reared cats is very much less than that of normal cats. A large moving visual stimulus is also less effective at producing a compensatory nystagmus than in normal cats, although some directions are followed better than others. There is little recovery within two months.

9. T. Bliss - Role of Monoamines in Hippocampal Long-Term Potentiation.

Noradrenergic pathways have been implicated in the "synaptic switching" which can be induced in the visual cortex during early post-natal life. We are investigating the possibility that the monoaminergic projections to hippocampus play a role in the functional synaptic plasticity represented by long-term potentiation of synaptic transmission.

10. R. Sutherland - Good to the Last Drop?

If thirsty rats drink novel, coffee-flavoured water and then receive LH stimulation, they subsequently prefer coffee over tap water. The relative magnitude of preferences induced by different durations of LH stimulation permits discrimination of the changes in the reinforcing effect which occur during a stimulation train.

11. P. Dodd - Temporal Relations as Discriminative Cues: Rate of Key Colour Alternation.

Two rates of key colour alternation were discriminative stimuli for pigeons in a two-component multiple schedule. Differences between response rates were small, but responding to a control key that allowed birds to shift freely between the two components showed good discriminative performance. Generalization tests on the rate dimension yielded sloping gradients and peak shift.

12. W. Honig - A Time for Decision: Evidence for Different Working Memory Processes in the Pigeon.

Pigeons could decide whether to respond to the second stimulus in a trial either on the basis of information contained in the initial stimulus (Delayed Discrimination), or on the basis of a conditional relationship between the two stimuli (Conditional Matching). The former procedure resulted in better performance when delays between the two stimuli were introduced.

13. M. Cynader & R. Douglas - Cortical Responses to Movement.

Visual cells may respond to a wide variety of stimuli and a satisfactory characterization of the responses of an individual unit should go beyond a description of its optimum stimulus. It is however uneconomical merely to list the unit's response to any of myriad possible stimuli. We have devised a cuboidal structure whose axes are space, space, and time to enable us to predict the responses of a given unit to arbitrary moving stimuli with different speeds, accelerations, and directions.

14. R. Klein - Explorations of Mental Size Scaling.

It has recently been demonstrated that subjects may gradually transform the size of the representation of a visual object when deciding if two differently sized stimuli have the same shape. Our experiments reveal the generality of this finding, and by testing its boundary conditions provide useful information about the nature of mental size scaling.

15. K. Beverley - Changing-Disparity and Changing-Size Inputs Feed the Same Motion-in-Depth Stage.

A sensation of motion in depth can be produced either by changing-disparity stimulation or by changing-size stimulation. I intend to report evidence that a changing-disparity stimulus and a changing-size stimulus feed the same motion-in-depth mechanism.

16. L. Balcom & M. Ozier - The Mental Representation of Spelling in Adults.

Pilot data are reported which show that recognition memory for spelling errors is specific for the spelling pattern presented. Judgments of meaning rather than spelling decreased spelling recognition accuracy significantly but only slightly in absolute terms. The data have relevance to our general knowledge about characteristics of the episodic memory trace.

17. J. Fentress - In Praise of Alchemy (or - Topsy Turvey).

I previously got into trouble with this term. But I assert that behavioral alchemy remains the zeitgeist for most experimental psychology (not just clinical). And its not all bad. However, non-equilibrium thermodynamics may be a more interesting model, with dynamic pattern and self-organization as focal points. And even mice do it!

18. F. Baker - The Interaction of Retinal and Tectal Cells of Goldfish in Tissue Culture.

The visual system in goldfish involves two populations of neural cells, those in the retina and those in the optic tectum. It is the interaction of these two populations *in vitro* (tissue culture), that I wish to talk about.

In culture retinal ganglion growth can be seen to enter pieces of tectal tissue but it is unclear what happens when this occurs without doing elaborate histological procedures. The difficulty in tissue culture arises in dissociating tectal cells, and then determining when a cell is living. Having made some progress in this, I wish to show some results from early experiments combining retinal tissue and single tectal cells *in vitro*.

19. D. Regan - Visually-Guided Locomotion: A Psychophysical Channel Sensitive to Flow Patterns in the Visual Field.

When the head moves forward, the stationary outside world appears to flow radially outwards from the point towards which the head is moving. Gibson proposed that this flow pattern is a stimulus for motion perception, and can be used to estimate one's direction of motion in three dimensions.

We previously reported evidence for psychophysical channels that were specifically sensitive to changing size. Orthogonality was the noteworthy property of these channels: they did not respond to motion or to flicker, even though these stimuli necessarily accompany changing size (Vision, Res., 18, 415, 1978). Properties of single units in cat parastriate cortex could provide a physiological basis for these channels (Regan & Cynader, ARVO, 1978). Here we report that visual sensitivity to changes in the size of a 0.5^{o} test square was depressed following 10 minutes inspection of a radially-flowing pattern of short annular line segments. This depression only occurred, however, when the square was located very close (about 0.3^{o}) to where the focus of the adapting flow pattern had been located. Hence, we propose that visual judgement of one's direction of forward motion may be mediated by changing-size channels. Since changing-size channels respond only to objects narrower than about 1.5^{o} (Beverley & Regan, Vision Res., in press), our proposal might go some way to explaining the precision of directional judgements of motion when driving an automobile or landing an aircraft.

20. J. Stevenson - Morphology of Tectal Reimplants in Goldfish.

Physiological recordings from excised and reimplanted portions of the optic tectum of goldfish have been used to study factors involved in the patterning of connections in the CNS of vertebrates. Unfortunately, the recording techniques employed have not permitted distinction between pre- and postsynaptic evoked potentials; the integrity of the reimplanted tissue may be questioned. I will present preliminary results of anatomical studies of such reimplanted pieces of tissue.

21. R. Brown - Pre-ejaculatory Ultrasonic Vocalizations of Male Rats.

Male rats emit 22 kHz pre-ejaculatory vocalizations when approaching females after they had achieved four or more ejaculations. The onset of these vocalizations was associated with increases in mounts without intromissions and decreases in intromission rate. On those ejaculatory series with 22 kHz pre-ejaculatory vocalizations, the females showed a lower lordosis intensity and a higher number of agonistic responses than on series with no vocalizations. It is suggested that the pre-ejaculatory vocalizations may serve to inhibit female agonistic behavior and stimulate sexual responsiveness.